

1. Literature Survey

Recent advancements in autonomous systems highlight the importance of accurate object tracking in dynamic environments. Sensor fusion techniques combine data from radar and lidar to improve tracking performance and reduce measurement uncertainty.

Traditional methods such as the Kalman Filter work well for linear systems but are not suitable for nonlinear real-world problems. The Extended Kalman Filter improves this by linearizing nonlinear models, but it introduces approximation errors and increases computational complexity.

To overcome these limitations, the Unscented Kalman Filter (UKF) is used. The UKF applies the Unscented Transform and sigma points to handle nonlinear systems more accurately without linearization, resulting in better estimation performance. Studies show that UKF-based sensor fusion improves tracking accuracy, reduces RMSE, and maintains consistent NIS values. It also supports reliable performance in radar-only, lidar-only, and combined sensor modes, making it suitable for real-time applications.

2. Introduction and Background

Autonomous systems rely heavily on accurate object tracking to ensure safe and efficient operation in dynamic environments. However, real-world tracking problems are inherently nonlinear and affected by noise, uncertainty, and sensor limitations. Traditional estimation techniques such as the Kalman Filter are limited to linear systems and fail to provide accurate results in nonlinear conditions.

To address these challenges, advanced filtering methods such as the Unscented Kalman Filter (UKF) have been developed. The UKF uses the Unscented Transform and sigma points to accurately propagate mean and covariance through nonlinear transformations, providing better estimation performance compared to Extended Kalman Filter (EKF) methods [1][2].

In autonomous tracking systems, sensor fusion techniques combine data from multiple sensors such as radar and lidar to improve accuracy and robustness. The UKF is

particularly effective in such systems due to its ability to handle nonlinear estimation without requiring linearization [3][4].

This literature survey focuses on the application of Unscented Kalman Filter-based sensor fusion techniques for object tracking, with emphasis on improving estimation accuracy, reducing RMSE, maintaining NIS consistency, and ensuring reliable performance across radar-only, lidar-only, and combined sensor modes.

3. Problem Statement

In autonomous tracking systems, accurately estimating the state of moving objects is challenging due to nonlinear system dynamics, sensor noise, and uncertainty in measurements. Traditional Kalman Filter methods are limited to linear systems, while Extended Kalman Filter (EKF) approaches introduce approximation errors due to linearization.

These limitations affect tracking accuracy, increase RMSE, and reduce consistency in NIS values, especially in dynamic environments. Additionally, maintaining reliable performance across different sensor modes such as radar-only, lidar-only, and combined fusion remains a challenge.

Therefore, there is a need for an advanced nonlinear estimation technique that can provide accurate state estimation, reduce error, and ensure consistent performance in real-time applications.

4. Research Objectives

- To analyze the performance of Unscented Kalman Filter for nonlinear object tracking.
- To implement a correct prediction and update process for radar and lidar measurements.
- To evaluate system performance under different modes such as radar-only, lidar-only, and combined sensor fusion.
- To minimize tracking error by reducing Root Mean Square Error (RMSE).
- To ensure consistency of Normalized Innovation Squared (NIS) values within the acceptable range.
- To improve accuracy and robustness of state estimation in dynamic and noisy environments.

5. Methodology of Literature Review

The literature survey follows a systematic approach to analyze research related to nonlinear state estimation and sensor fusion techniques. Relevant research papers were collected from reliable sources such as IEEE Xplore, Springer, ScienceDirect, and MDPI databases [2][8].

The selection of papers was based on keywords including Unscented Kalman Filter, sensor fusion, nonlinear tracking, RMSE, NIS, radar tracking, and lidar-based estimation. Only studies focusing on nonlinear estimation methods and real-time tracking applications were considered [4].

A total of 10 research papers were selected for detailed analysis. The selected studies were evaluated based on performance metrics such as estimation accuracy, RMSE reduction, NIS consistency, computational efficiency, and real-time applicability [3].

This structured methodology ensures a clear and reliable understanding of existing techniques and helps identify limitations in current approaches.

6. Review of Existing Work

Recent research in autonomous systems has focused on improving object tracking performance using advanced nonlinear estimation techniques. Sensor fusion approaches that combine radar and lidar data have been widely adopted to enhance tracking accuracy and robustness in dynamic environments [2][5].

The Kalman Filter has been extensively used for state estimation due to its simplicity and computational efficiency. However, it is limited to linear systems and performs poorly when applied to nonlinear tracking problems [4]. To overcome this limitation, the Extended Kalman Filter (EKF) was introduced, which approximates nonlinear models using linearization. Although EKF improves performance, it introduces errors due to Jacobian-based approximations and requires careful parameter tuning [8].

To address these challenges, the Unscented Kalman Filter (UKF) has been proposed as a more accurate nonlinear estimation method. The UKF uses sigma points and the Unscented Transform to capture the mean and covariance of nonlinear systems without linearization, resulting in improved estimation accuracy [1][3].

Several studies demonstrate that UKF-based sensor fusion significantly reduces tracking error by minimizing RMSE and maintaining consistency in NIS values. Compared to EKF, the UKF provides better performance in highly nonlinear environments and improves robustness against sensor noise and uncertainties [6][7].

However, the UKF introduces higher computational complexity due to the generation and propagation of sigma points. This creates a trade-off between estimation accuracy and computational efficiency, especially in real-time applications involving multiple object tracking.

Overall, existing research indicates that UKF-based sensor fusion provides a more reliable and accurate solution for nonlinear object tracking, but challenges related to computational cost and real-time implementation still remain.

7. Critical Analysis and Comparison

The Kalman Filter is efficient but limited to linear systems, making it unsuitable for nonlinear tracking problems [4]. The Extended Kalman Filter improves performance but introduces approximation errors due to linearization [8].

The Unscented Kalman Filter (UKF) provides better accuracy by handling nonlinear systems without linearization, resulting in improved RMSE and NIS performance [1][3]. Sensor fusion using radar and lidar further enhances tracking accuracy, but increases computational complexity [2][5].

Method	Advantages	Limitations
Kalman Filter	Fast, simple, low computation	Only for linear systems
Extended Kalman Filter	Handles nonlinear systems	Linearization error, complex tuning
Unscented Kalman Filter (UKF)	High accuracy, no linearization, better RMSE & NIS	Higher computational cost
Sensor Fusion (Radar + Lidar)	Improved accuracy, robust tracking	Complex integration, higher cost

The UKF offers better accuracy and robustness for nonlinear tracking compared to EKF, but requires higher computational resources.

8. Research Gaps

Despite improvements in nonlinear tracking methods, several limitations still exist. Many studies focus on improving accuracy but do not fully address real-time computational efficiency in UKF-based systems. Additionally, maintaining consistent NIS values and minimizing RMSE under highly dynamic conditions remains a challenge.

Furthermore, limited research has been conducted on optimizing UKF performance across different sensor modes such as radar-only, lidar-only, and combined fusion. These gaps highlight the need for a more efficient and robust tracking approach.

9. Alignment to Proposed Work / Project

The reviewed literature provides a strong foundation for implementing the Unscented Kalman Filter for object tracking. The identified limitations, such as nonlinear estimation challenges, RMSE reduction, and NIS consistency, directly guide the design of this project.

The proposed system focuses on applying UKF-based sensor fusion using radar and lidar data to improve tracking accuracy. It also evaluates performance under radar-only, lidar-only, and combined modes, ensuring reliable and consistent results.

Thus, the insights from the literature are directly used to develop an efficient and robust tracking system that meets real-time performance requirements.

10. Concepts / Rules / Principles / Laws

10.1 Concepts:

- Nonlinear estimation
- Sigma points
- Sensor fusion
- State prediction
- Measurement update

10.2 Rules:

- No future data usage
- Sequential processing
- Consistent updates
- Noise consideration
- Real-time execution

10.3 Principles:

- Uncertainty propagation
- Statistical estimation
- Data fusion
- Nonlinear transformation
- Error minimization

10.4 Laws:

- Gaussian distribution
- Bayesian inference
- Probability theory
- State-space modeling
- Covariance propagation

11. System Model and State Space Representation

In object tracking, the system is represented using a state-space model that describes the evolution of the system state and its observations over time. The state vector typically includes position and velocity components, which are updated using nonlinear process and measurement models [4].

The Unscented Kalman Filter uses this state-space representation to predict and update the system state based on incoming radar and lidar measurements, enabling accurate estimation in nonlinear environments [1][3].

12. What-if Analysis

What if the system becomes highly nonlinear?

The Unscented Kalman Filter provides better accuracy by handling nonlinear transformations without linearization.

What if sensor noise increases?

Higher noise reduces estimation accuracy and increases RMSE, affecting overall tracking performance.

What if only radar data is used?

Radar provides good velocity information but may have lower position accuracy compared to lidar.

What if only lidar data is used?

Lidar gives accurate position measurements but lacks direct velocity information.

What if both sensors are fused?

Sensor fusion improves accuracy and robustness by combining complementary information from both sensors.

What if RMSE is high?

A high RMSE indicates poor estimation performance and reduced tracking accuracy.

What if NIS values are outside the range?

It shows that the filter is inconsistent and not properly tuned for the system.

What if sigma points are not properly selected?

The UKF may fail to accurately capture nonlinear behavior, leading to incorrect state estimation.

What if process noise is very low?

The filter becomes less responsive to changes in system dynamics.

What if process noise is very high?

The estimation becomes unstable and less reliable.

13. Tools, Technologies, and Methods Used

Tools:

- MATLAB
- Python
- Simulation environment

Technologies:

- Sensor fusion

- Radar sensing
- Lidar sensing
- Autonomous systems

Methods:

- Unscented Kalman Filter (UKF)
- State prediction
- Measurement update
- RMSE evaluation
- NIS consistency check

14. Conclusion

This literature survey analyzed various object tracking techniques with a focus on nonlinear estimation methods. The study highlighted the limitations of Kalman Filter and Extended Kalman Filter, and the advantages of the Unscented Kalman Filter for handling nonlinear systems.

The results show that UKF-based sensor fusion improves tracking accuracy, reduces RMSE, and maintains NIS consistency across different sensor modes. These findings support the development of a reliable and efficient real-time tracking system.

Reference

1. [Unscented Kalman Filter \(UKF\) - Sensor Fusion](#)
2. [Adaptive Robust Unscented Kalman Filter for AUV Acoustic Navigation - PMC](#)
3. [Radar-Based Target Tracking Using Deep Learning Approaches with Unscented Kalman Filter](#)
4. [A Robust Adaptive Unscented Kalman Filter for Floating Doppler Wind-LiDAR Motion Correction](#)
5. [1908.07604](#)

6. [2002.00878](#)
7. [Unscented Kalman Filter Tutorial - The Institute for Computational ...](#)
8. [A Robust Adaptive Unscented Kalman Filter for Floating Doppler Wind-LiDAR Motion Correction](#)
9. [Radar-Based Target Tracking Using Deep Learning Approaches with Unscented Kalman Filter](#)
10. [Adaptive Robust Unscented Kalman Filter for AUV Acoustic Navigation - PMC](#)
11. [Unscented Kalman Filter \(UKF\) - Sensor Fusion](#)
12. [Unscented Kalman Filter Tutorial download / unscented-kalman-filter-tutorial.pdf / PDF4PRO](#)
13. [1802.01235](#)

